



SoMe Profiles

[in linkedin.com/in/peter-buch-hansen](https://www.linkedin.com/in/peter-buch-hansen)

github.com/peterbuchhansen

Languages

Danish Native

English C2 / Proficiency

German B1 / Intermediate

Certifications

Forklift certificate · AMU, 2023.

Skills

DevOps & Tooling

Docker, Podman, Ansible, Kubernetes, Proxmox, CI/CD, Drone CI, Artifactory, Bazel, CMake, Git, DevContainers, Bash, Docker, Podman, Ansible, Kubernetes, Proxmox, CI/CD, Drone CI, Artifactory, Bazel, CMake, Git, DevContainers, Bash

Robotics

PCB design, ROS, AprilTag, camera calibration, motor control / sensors

Databases

PostgreSQL, MySQL, SQLite, schema design & DTOs, SQL

Programming

Rust, C++, Python, TypeScript, JavaScript, Java, HTML/CSS, Bash, Structured Text (PLC), MATLAB, Jsonnet

Awards

MiR Hackathon Q1 2024 · Intra tool.

(Mass upgrade for robots/simulators)

MiR Hackathon Q1 2023 · Simsons.

(DevOps robot simulation tool)

References

Benjamin Jensen +45 2071 0538

[linkedin.com/in/benjamin-jensen-dk](https://www.linkedin.com/in/benjamin-jensen-dk)

Was My Manager @ MiR 2021-2024

Currently Manager @ GF Insurance

Peter Buch Hansen

Software Developer & Robotics Engineer · Project, Tech & Team Lead

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[Complete Resume available at peterbuchhansen.github.io/Resumify](https://peterbuchhansen.github.io/Resumify)

Summary

Software & Robotics Engineer focused on system design, automation, and real-time data analysis. Led MiR's RAT platform from prototype to a production-ready system for global robots and fleet operations. Strong in Python, C++, Rust, TypeScript, HTML/CSS, Bash; comfortable with MVC architectures, URCap/HMI, and 2D/3D sensor analysis. Heavy use of REST, gRPC, WebSockets, pub/sub, and MySQL, SQLite, PostgreSQL.

Key achievements

- Migrated core RAT services Python → Rust for robustness and performance.
- Modernized RAT UI (Angular / Bootstrap) replacing RESTool.
- 98% uptime during major platform rewrites.
- Led cross-functional teams (engineers, consultants, students).
- Delivered a scalable automated testing platform in CI/CD, production, and R&D.

Experience

Mobile Industrial Robots

Jan 2024 — Present · (4 yr 10 mos)

Project Lead & System Architect / Software Developer
Odense, Denmark · Robot Automation Test (RAT) platform.

- Led MiR's RAT platform; modernized architecture, scaled automation, high reliability across production, R&D, and test sites.
- Rewrote core platform Python → Rust; OpenAPI (FastAPI) with SSO; gRPC, REST, WebSockets.
- Built and led a team of 5 (engineers, consultants, students).
- 98% uptime through continuous development and deployment.

Mobile Industrial Robots

Jan 2023 — Jan 2024

Team Lead & Tech Lead / Software Developer
Scaled and stabilized RAT for R&D, data collection, and multi-robot testing.

- Architecture evolution from ROS modules toward gRPC microservices.
- Integrations with robots, test PCBs, CI/CD, Artifactory; deployments with Ansible.
- ReportPortal for centralized test analytics; maintained coverage through refactors.

Mobile Industrial Robots

Jul 2022 — Jan 2023

Project Lead & Robot Automation Test Engineer
Owned RAT MVP as project lead and sole developer; scaled for production, R&D, and test.

- Single-handedly developed and operated RAT for automation and data collection.
- Restructured codebase and infrastructure for stability and performance.
- Partnered with QA, R&D, and production for reliable operations.

Mobile Industrial Robots

Jan 2022 — Jul 2022

Robot Automation Test Engineer
Full-time R&D after graduation; ROS- and Docker-based test systems.

- Expanded RAT proof-of-concept; integrated sensors and platforms with QA and hardware teams.
- Documentation and version control via Artifactory.

Mobile Industrial Robots

Jun 2021 — Jan 2022 · 8 mos

Student worker & B.Eng project

Pose-estimation camera system for automated test area (AprilTags, multi-worker camera stack, Docker/ROS, MATLAB, DBSCAN, Charuco / Zhang calibration).

Mobile Industrial Robots

Feb 2021 — Jun 2021

Intern, R&D

Test automation in Python/C++ with ROS; CI/CD (CMake, Docker, Artifactory, Drone); PCB module for automated test.

Freelance

2012 — 2018

Web developer

Websites for volunteer organizations (e.g. Inca Spejderne Sydfyn, BUS Svendborg).

Education

University of Southern Denmark

Sep 2018 — Jan 2022

B.Eng Robot Systems

sdu.dk — Robot Systems. Semester projects: PLC & motion, custom UR-Cap & gripper, vision-based ball pickup (OpenCV, UR), Siemens inverted pendulum, MVC web app (MySQL), embedded micro-robot.